

The Hybrid Density Matrix Framework: A Diagnostic Architecture for Reliability Assessment in NISQ Quantum Systems

Michael Aspinall
*Independent Researcher**
(Dated: June 27, 2026)

Quantum computing in the Noisy Intermediate-Scale Quantum (NISQ) era is fundamentally constrained by decoherence, gate imperfections, measurement errors, and environmental noise. While density matrices provide complete descriptions of quantum states, and quantum channels model noise evolution, practical quantum computing demands more than state characterisation alone: researchers must assess how *trustworthy* a computation is, not merely how accurate the final state appears to be.

This paper introduces the **Hybrid Density Matrix Framework** (HDMF), a structured diagnostic architecture that augments conventional density matrix representations with layered information describing noise, uncertainty, hardware context, and measurement history. The framework is formalised as

$$\mathcal{H}_\rho = (\rho, \mathcal{N}, \mathcal{U}, \mathcal{C}, \mathcal{M}),$$

where each auxiliary layer carries explicit semantics and a well-defined update rule, preserving standard quantum mechanics while adding operationally grounded diagnostic metadata.

To avoid double-penalising correlated degradation, we introduce a *decoupled* reliability metric

$$R_{\text{HDM}} = \mathcal{F} \cdot \gamma_\perp \cdot \mathcal{S},$$

where $\gamma_\perp$ is the *fidelity-residual purity*—the component of state purity not already explained by fidelity loss. This construction satisfies boundedness, monotonicity, and failure sensitivity while eliminating the information-overlap problem present in the naive product $F \cdot \gamma \cdot S$.

We identify and formally characterise the **Hidden Reliability Region**: the set of quantum states exhibiting high fidelity yet low operational reliability, a regime largely invisible to conventional metrics. Simulation over 31,000 trials, evaluated against a Bell-state target under mixed NISQ noise (amplitude damping, dephasing, readout error), demonstrates that R_{HDM} predicts downstream computational fidelity with correlation $r = 0.9717$, versus $r \approx 0$ for checkpoint fidelity alone. We further identify boundary conditions under which simpler metrics—notably purity—are sufficient, providing honest guidance for practitioners.

The framework is designed to complement, not replace, established techniques including randomised benchmarking, quantum error mitigation, and Bayesian state estimation. Physical hardware validation and information-theoretic grounding of the reliability axiom are identified as the principal directions for future work.

PACS numbers: 03.67.Pp, 03.67.Lx, 03.65.Yz

I. INTRODUCTION

The success of a quantum computation depends not only on preparing a target quantum state but on determining whether that state remains *trustworthy* throughout the computation. This distinction—between state quality and operational trustworthiness—is central to the reliability challenge in NISQ-era quantum systems [1].

Reliability assessment is already an active research area, encompassing hardware benchmarking [3, 4], quantum state estimation [6], verification protocols, and error-mitigation techniques [5]. The Hybrid Density Matrix Framework does not introduce reliability as a new concept. Rather, it provides a structured diagnostic architecture that integrates multiple reliability-relevant signals—currently scattered across distinct methodologies—into

a unified framework for operational trust assessment in noisy quantum systems.

Traditional density matrix formalism provides a mathematically complete description of mixed quantum states. Kraus operators and Lindblad master equations provide robust frameworks for modelling open quantum systems [2]. However, modern NISQ hardware introduces a practical challenge: researchers are rarely interested only in the state itself. They also need to understand:

- How much noise influenced the state.
- Whether measured outcomes remain reliable.
- How uncertainty evolves throughout computation.
- Whether observed fidelity accurately reflects operational trustworthiness.

This motivates a diagnostic extension to the density matrix formalism.

* available via GitHub repository

A. Primary Contributions

The main contributions of this work are:

1. **Formalised framework with explicit semantics.** We define $\mathcal{H}_\rho = (\rho, \mathcal{N}, \mathcal{U}, \mathcal{C}, \mathcal{M})$ with explicit update rules for each auxiliary layer (Section II), addressing the ambiguity present in earlier informal treatments.
2. **Decoupled reliability metric.** We introduce $R_{\text{HDM}} = \mathcal{F} \cdot \gamma_\perp \cdot \mathcal{S}$ with the fidelity-residual purity $\gamma_\perp$ (Section III), resolving the information-overlap problem in the naive multiplicative form.
3. **Hidden Reliability Region.** We formally define and empirically demonstrate a regime where high fidelity coexists with low operational trustworthiness (Section IV), with testable predictions for physical hardware.
4. **Empirical validation.** Simulation across 31,000 trials demonstrates that R_{HDM} predicts downstream performance ($r = 0.9717$) far more effectively than fidelity alone ($r \approx 0$), with honest characterisation of boundary conditions (Section VII).
5. **Integration architecture.** We demonstrate how the HDMF complements—rather than replaces—randomised benchmarking, error mitigation, and Bayesian state estimation (Section V).

II. FORMAL DEFINITION OF THE HYBRID DENSITY MATRIX FRAMEWORK

A. Motivation

Consider two quantum states ρ_A and ρ_B with identical measured fidelity $\mathcal{F}(\rho_{\text{ideal}}, \rho_A) = \mathcal{F}(\rho_{\text{ideal}}, \rho_B) = 0.95$ with respect to an ideal target. Traditional analysis concludes both are equally successful. However, suppose ρ_A arises from a highly stable, well-calibrated circuit, while ρ_B emerges from a heavily degraded system experiencing significant decoherence, calibration drift, and accumulated gate error. The density matrix itself does not preserve this distinction.

The HDMF addresses this limitation by augmenting the state representation with structured diagnostic layers that characterise reliability, uncertainty, noise, and context—a *reliability envelope* surrounding the conventional density matrix.

B. Formal Definition

Definition 1 (Hybrid Density Matrix). *A Hybrid Density Matrix is the tuple*

$$\mathcal{H}_\rho = (\rho, \mathcal{N}, \mathcal{U}, \mathcal{C}, \mathcal{M}), \quad (1)$$

where:

- $\rho \in \mathbb{C}^{d \times d}$ is the physical density matrix.
- $\mathcal{N}$ is the noise descriptor layer.
- $\mathcal{U}$ is the uncertainty quantification layer.
- $\mathcal{C}$ is the classical hardware context layer.
- $\mathcal{M}$ is the measurement and mitigation history layer.

The density matrix ρ is the sole representation of the physical quantum state. The auxiliary layers provide diagnostic information and do not alter the underlying quantum mechanical interpretation of ρ .

C. State Layer

The state layer is the conventional density matrix $\rho \in \mathbb{C}^{d \times d}$ satisfying:

$$\rho = \rho^\dagger, \quad \rho \succeq 0, \quad \text{Tr}(\rho) = 1. \quad (2)$$

This layer preserves all standard quantum mechanical observables, expectation values, entanglement properties, and state reconstruction techniques. No modifications are made to the density matrix formalism itself.

D. Noise Descriptor Layer

The noise descriptor layer captures known or inferred mechanisms responsible for state degradation:

$$\mathcal{N} = \{(\mathcal{E}_i, p_i)\}_{i=1}^k, \quad (3)$$

where each $\mathcal{E}_i$ is a completely positive trace-preserving (CPTP) map [2] and $p_i \in [0, 1]$ is the associated error probability or rate. The framework is agnostic regarding noise representation: Kraus operators, Lindblad generators, calibrated hardware error rates, or empirical error models may all populate $\mathcal{N}$.

Update rule. After each gate operation $\mathcal{E}_j$ is applied:

$$\mathcal{N} \leftarrow \mathcal{N} \cup \{(\mathcal{E}_j, p_j)\}, \quad \rho \leftarrow \mathcal{E}_j(\rho). \quad (4)$$

E. Uncertainty Quantification Layer

The uncertainty layer captures confidence in the reconstructed state and associated measurements:

$$\mathcal{U} = (\mathcal{U}_{\text{local}}, \mathcal{U}_{\text{global}}, \mathcal{U}_{\text{op}}), \quad (5)$$

with the following components.

Local uncertainty. Characterises confidence in individual matrix elements:

$$\mathcal{U}_{\text{local}} = \Sigma_\rho, \quad (6)$$

where Σ_ρ is the covariance structure associated with estimated state parameters, obtainable from quantum state tomography [6].

Global uncertainty. Represents confidence in the state reconstruction as a whole:

$$\mathcal{U}_{\text{global}} = P(\rho \mid \mathcal{D}), \quad (7)$$

where $\mathcal{D}$ denotes measurement data. In Bayesian implementations, this is a posterior distribution over admissible density matrices.

Operational uncertainty. A scalar confidence estimate:

$$\mathcal{U}_{\text{op}} = R_{\text{HDM}} \in [0, 1], \quad (8)$$

defined in Section III.

Update rule. After n_{new} additional measurements are acquired and state tomography is updated:

$$\Sigma_\rho \leftarrow (\Sigma_\rho^{-1} + n_{\text{new}} I_{\text{Fisher}})^{-1}, \quad (9)$$

where I_{Fisher} is the quantum Fisher information matrix for the measurement basis employed.

F. Classical Hardware Context Layer

The context layer stores classical variables describing the computational environment:

$$\mathcal{C} = \{(k_j, v_j)\}_{j=1}^m, \quad (10)$$

where each entry is a key-value pair. Potential entries include calibration timestamps, device temperature, pulse configurations, gate schedules, hardware topology, qubit assignment maps, drift measurements, and randomised benchmarking results.

Update rule. Context entries are updated upon hardware recalibration, between circuit executions, or whenever a diagnostically significant hardware event occurs.

G. Measurement and Mitigation History Layer

The measurement layer records historical information associated with state preparation and correction procedures:

$$\mathcal{M} = \{(t_\ell, \mathcal{P}_\ell, r_\ell)\}_{\ell=1}^s, \quad (11)$$

where t_ℓ is a timestamp, $\mathcal{P}_\ell$ is the protocol identifier (tomography, error mitigation, benchmarking, *etc.*), and r_ℓ is the result record. Unlike the state layer, which represents the current quantum state, $\mathcal{M}$ preserves the *path* by which that state was obtained, enabling retrospective reliability auditing.

Update rule. $\mathcal{M}$ is append-only: each measurement round, mitigation step, or benchmarking run appends a new record. No record is ever deleted.

H. Framework Evolution

The HDM evolves through a computation as:

$$\mathcal{H}_\rho(t+1) = (\mathcal{E}(\rho_t), \mathcal{N}'_t, \mathcal{U}'_t, \mathcal{C}'_t, \mathcal{M}'_t), \quad (12)$$

where the state layer evolves through $\rho_{t+1} = \mathcal{E}(\rho_t)$ and each auxiliary layer updates according to the rules defined above (Eqs. (4), (9), and the context and history update rules). The evolution is Markovian for the state layer; the auxiliary layers may retain full history.

I. Assumptions and Scope

- A1.:** The standard density matrix formalism and its physical interpretation remain unchanged.
- A2.:** Reliability, as defined in this work, is an engineering and operational construct—not a new fundamental quantum observable.
- A3.:** Hardware survival probabilities are estimated under the assumption of independent gate failures. This assumption may not hold under strongly correlated noise (see Section VIII).
- A4.:** Empirical validation is performed via classical simulation. Validation on physical NISQ hardware is left for future work.
- A5.:** The multiplicative form of R_{HDM} is one valid realisation of the Reliability Axiom; alternative aggregation functions are discussed in Section III H.

III. RELIABILITY AXIOM AND METRIC CONSTRUCTION

A. Motivation: The Information-Overlap Problem

The most natural reliability metric integrating fidelity, purity, and hardware survival is the product $F \cdot \text{Tr}(\rho^2) \cdot \prod_i (1 - e_i)$. However, this formulation suffers from an *information-overlap problem*: both fidelity and purity decrease when the state undergoes decoherence, so their product double-penalises the same underlying degradation.

To see this, note that for a state undergoing depolarising noise at rate p :

$$\mathcal{F} = 1 - \frac{3}{4}p + O(p^2), \quad (13)$$

$$\text{Tr}(\rho^2) = 1 - \frac{3}{2}p + O(p^2). \quad (14)$$

The product $F \cdot \text{Tr}(\rho^2)$ penalises p at rate $\approx \frac{9}{4}$ to first order, whereas the true reliability loss due to the noise is only linear in p . This inflation of the penalty is an artefact of correlated inputs, not a reflection of independent failure modes.

B. Fidelity-Residual Purity

We resolve this by decomposing purity into a component explained by fidelity loss and an independent residual:

Definition 2 (Fidelity-Residual Purity). *Let ρ_{obs} be an observed state with fidelity $\mathcal{F} = \mathcal{F}(\rho_{\text{ideal}}, \rho_{\text{obs}})$ and purity $\gamma = \text{Tr}(\rho_{\text{obs}}^2)$. The fidelity-predicted purity under depolarising noise is*

$$\hat{\gamma}(\mathcal{F}) = 2\mathcal{F}^2 - 1, \quad (15)$$

obtained from the relation $\mathcal{F} = \frac{1+\sqrt{2\gamma-1}}{2}$ for pure-to-mixed depolarisation. The fidelity-residual purity is

$$\gamma_{\perp} = \frac{\gamma}{\hat{\gamma}(\mathcal{F})} = \frac{\text{Tr}(\rho_{\text{obs}}^2)}{2\mathcal{F}^2 - 1}, \quad (16)$$

clipped to $[0, 1]$. When $\gamma_{\perp} < 1$, the state is less pure than fidelity alone would predict—indicating additional coherence loss not captured by state similarity.

Remark 1. Equation (15) is derived for the depolarising channel. For other noise models, $\hat{\gamma}(\mathcal{F})$ should be computed from the model-specific fidelity-purity relation. In the absence of a calibrated model, the conservative choice $\hat{\gamma}(\mathcal{F}) = \mathcal{F}$ (so that $\gamma_{\perp} = \gamma/\mathcal{F}$) is recommended.

C. Reliability Axiom

Axiom 1 (Operational Reliability). *A quantum computation should be considered operationally reliable only if it simultaneously satisfies acceptable levels of state accuracy ($\mathcal{F}$), structural integrity beyond what fidelity predicts ($\gamma_{\perp}$), and hardware viability ($\mathcal{S}$). Failure in any single dimension reduces the trustworthiness of the overall computation for downstream decision-making.*

This principle reflects the weakest-link behaviour common in reliability engineering and fault-sensitive computational workflows.

D. Metric Requirements

Let $\mathcal{F}$ denote fidelity, $\gamma_{\perp}$ fidelity-residual purity, and $\mathcal{S}$ hardware survival probability. A reliability metric $R(\mathcal{F}, \gamma_{\perp}, \mathcal{S})$ should satisfy:

Boundedness:: $R \in [0, 1]$.

Normalisation:: $R(1, 1, 1) = 1$.

Monotonicity:: $\partial R / \partial \mathcal{F} > 0$, $\partial R / \partial \gamma_{\perp} > 0$, $\partial R / \partial \mathcal{S} > 0$.

Failure sensitivity:: $R \rightarrow 0$ whenever $\mathcal{F} \rightarrow 0$, $\gamma_{\perp} \rightarrow 0$, or $\mathcal{S} \rightarrow 0$.

Decoupling:: the input dimensions are informationally independent.

The fifth requirement is what distinguishes the present metric from the naive product $\mathcal{F} \cdot \text{Tr}(\rho^2) \cdot \mathcal{S}$.

E. Decoupled Multiplicative Reliability Metric

Definition 3 (HDM Reliability Score). *The HDM reliability score is:*

$$R_{\text{HDM}} = \mathcal{F}(\rho_{\text{ideal}}, \rho_{\text{obs}}) \cdot \gamma_{\perp}(\rho_{\text{obs}}) \cdot \mathcal{S}(\mathcal{N}), \quad (17)$$

where

$$\mathcal{S}(\mathcal{N}) = \prod_{(\mathcal{E}_i, p_i) \in \mathcal{N}} (1 - p_i), \quad (18)$$

and $\gamma_{\perp}$ is defined in Definition 2.

Proposition 1. *The metric R_{HDM} defined in Eq. (17) satisfies boundedness, normalisation, monotonicity, failure sensitivity, and the decoupling requirement.*

Proof sketch. Boundedness follows from $\mathcal{F}, \gamma_{\perp}, \mathcal{S} \in [0, 1]$. Normalisation holds when $\rho_{\text{obs}} = \rho_{\text{ideal}}$ (a pure state with $\gamma = 1 = \mathcal{F}$, giving $\gamma_{\perp} = 1$) and all error rates are zero. Monotonicity follows from $\partial(\mathcal{F} \cdot \gamma_{\perp} \cdot \mathcal{S}) / \partial x > 0$ for each $x \in \{\mathcal{F}, \gamma_{\perp}, \mathcal{S}\}$ when the others are fixed positive. Failure sensitivity follows from the multiplicative structure. Decoupling holds by construction: $\gamma_{\perp}$ is the component of purity not explained by $\mathcal{F}$, so $\text{Cov}(\mathcal{F}, \gamma_{\perp}) \approx 0$ under the depolarising model. $\square$

F. Interpretation

R_{HDM} combines three distinct and informationally non-redundant dimensions:

- **State accuracy ($\mathcal{F}$):** How close is the observed state to the target?
- **Residual structural integrity ($\gamma_{\perp}$):** Does the state exhibit additional coherence loss beyond what fidelity already signals?
- **Hardware viability ($\mathcal{S}$):** How reliable was the hardware pathway?

The multiplicative structure encodes the weakest-link principle: degradation in any single dimension reduces the overall confidence score, regardless of the other dimensions.

G. Illustrative Example

Consider two observed states where $\mathcal{F}_A = \mathcal{F}_B = 0.95$. Suppose:

$$\begin{aligned} \gamma_{\perp, A} &= 0.96, & \mathcal{S}_A &= 0.98, \\ \gamma_{\perp, B} &= 0.68, & \mathcal{S}_B &= 0.40. \end{aligned}$$

Then:

$$\begin{aligned} R_{\text{HDM}}^A &= 0.95 \times 0.96 \times 0.98 \approx 0.894, \\ R_{\text{HDM}}^B &= 0.95 \times 0.68 \times 0.40 \approx 0.258. \end{aligned}$$

Although fidelity is identical, the operational trustworthiness differs by a factor of ≈ 3.5 , a distinction invisible to fidelity alone.

H. Alternative Aggregation Functions

The Reliability Axiom does not uniquely determine the multiplicative form. Several alternatives satisfy boundedness and monotonicity:

$$R_G = (\mathcal{F} \gamma_\perp \mathcal{S})^{1/3}, \quad (19)$$

$$R_W = a\mathcal{F} + b\gamma_\perp + c\mathcal{S}, \quad a + b + c = 1. \quad (20)$$

These alternatives exhibit weaker failure sensitivity than the multiplicative form: under R_W , overall reliability remains elevated even when one component approaches zero. The multiplicative form is adopted here because it directly implements the weakest-link principle and provides interpretable behaviour for diagnostic use. Future work should compare these formulations empirically.

I. Reliability Regimes

Several qualitatively distinct operating regimes emerge under R_{HDM} :

TABLE I. Reliability regimes and their characteristics.

Regime	Description	Characteristics
I	High Reliability	$\mathcal{F} \approx 1, \gamma_\perp \approx 1, \mathcal{S} \approx 1$
II	Hidden Degradation	$\mathcal{F} \approx 1, \gamma_\perp \ll 1$
III	Hardware-Limited	$\mathcal{F} \approx 1, \gamma_\perp \approx 1, \mathcal{S} \ll 1$
IV	Complete Failure	$\mathcal{F} \rightarrow 0, \gamma_\perp \rightarrow 1/d, \mathcal{S} \rightarrow 0$

IV. THE HIDDEN RELIABILITY REGION

A. State Similarity Versus Operational Reliability

A fundamental assumption underlying current NISQ practice is that high fidelity implies successful computation. We challenge this assumption by identifying a regime—the Hidden Reliability Region—where the two quantities significantly decouple.

- Fidelity answers: “Where is the state?”
- Reliability answers: “How much should we trust the state?”

B. Formal Definition

Definition 4 (Hidden Reliability Region). Let $\tau_F \in (0, 1)$ be a fidelity threshold and $\tau_R \in (0, 1)$ be a reliability

threshold with $\tau_R < \tau_F$. The Hidden Reliability Region is the set of quantum states:

$$\mathcal{H} = \{\rho_{\text{obs}} \mid \mathcal{F}(\rho_{\text{ideal}}, \rho_{\text{obs}}) \geq \tau_F \wedge R_{\text{HDM}}(\rho_{\text{obs}}) < \tau_R\}. \quad (21)$$

States in $\mathcal{H}$ appear successful under conventional fidelity monitoring but carry low operational trustworthiness. They represent a source of false confidence in NISQ computations.

C. Why the Region Emerges

Fidelity may decay slowly while coherence ($\gamma_\perp$) and hardware confidence ($\mathcal{S}$) decay rapidly. The physical mechanism is as follows. Under mixed NISQ noise, the off-diagonal density matrix elements (coherences) are suppressed by dephasing and amplitude damping at rates often faster than the diagonal population shifts that reduce fidelity. Consequently, $\mathcal{F}(t)$ and $R_{\text{HDM}}(t)$ decouple, creating the regime in Eq. (21).

D. Growth of the Hidden Reliability Region

Numerical exploration suggests that $\mathcal{H}$ expands as circuit depth and noise exposure increase. Mixed NISQ noise channels incorporating decoherence, readout error, and calibration uncertainty produce larger Hidden Reliability Regions than simple depolarising models. This occurs because fidelity under depolarising noise decays monotonically with the noise parameter, whereas under mixed noise the relationship between fidelity and operational reliability is less regular, allowing greater divergence.

E. Testable Predictions

The Hidden Reliability Region yields the following testable predictions for future physical hardware experiments:

1. Bell-state experiments on physical NISQ hardware will exhibit parametric regions where fidelity remains ≥ 0.90 while $R_{\text{HDM}} < 0.60$.
2. The region will be more pronounced under mixed NISQ noise than under pure depolarising noise.
3. Longer circuits will enter the region at lower noise rates.
4. Error-mitigation methods that improve fidelity without improving purity (e.g. measurement error mitigation without error correction) will leave states within $\mathcal{H}$.

V. RELATIONSHIP TO ESTABLISHED TECHNIQUES

The HDMF is proposed as a reliability-oriented integration architecture. Table II summarises the primary question each technique addresses.

TABLE II. Comparison of reliability-related techniques.

Technique	Primary Question
State fidelity / tomography	How close is the state to ideal?
Randomised benchmarking	How reliable is the hardware?
Bayesian state estimation	What is the state, with what confidence?
Quantum error mitigation	Can noise effects be reduced?
HDMF	How much should we trust this computation overall?

A. Randomised Benchmarking

Randomised benchmarking (RB) [3, 4] provides scalable estimates of average gate error rates. These error rates naturally populate the noise descriptor layer $\mathcal{N}$ and contribute to the hardware survival component $\mathcal{S}$. The HDMF therefore treats RB as an upstream data source rather than a competing framework.

B. Quantum Error Mitigation

Techniques such as zero-noise extrapolation and probabilistic error cancellation [5] aim to improve output quality under realistic NISQ conditions. Within the HDMF, these approaches may be evaluated not only by their effect on $\mathcal{F}$ but also by their effect on $\gamma_{\perp}$ and $\mathcal{S}$ —a broader assessment than fidelity improvement alone. This distinction is important because an error-mitigation strategy may improve fidelity while leaving $\gamma_{\perp}$ largely unchanged, keeping the state within the Hidden Reliability Region.

C. Bayesian Quantum State Estimation

Bayesian quantum state estimation [6] provides principled uncertainty quantification for reconstructed quantum states. Such approaches align naturally with the uncertainty layer $\mathcal{U}$ and may serve as rigorous implementations of the global uncertainty component. In the most complete realisation of the framework, posterior state distributions may be incorporated directly into $\mathcal{U}_{\text{global}}$.

VI. BENCHMARK APPLICATIONS

A. Bell-State Benchmark

The target state is the Bell state:

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}, \quad (22)$$

constructed via $H(q_0)$ followed by $\text{CNOT}(q_0, q_1)$. The ideal density matrix is $\rho_{\text{ideal}} = |\Phi^+\rangle\langle\Phi^+|$, with

$$\rho_{\text{ideal}} = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}. \quad (23)$$

The off-diagonal terms $\rho_{14} = \rho_{41} = 0.5$ represent the coherence responsible for entanglement and are particularly sensitive to dephasing noise.

1. Baseline Noise Model

A depolarising channel serves as a control condition:

$$\mathcal{E}_{\text{dep}}(\rho) = (1-p)\rho + p\frac{\mathbb{I}}{4}, \quad (24)$$

where p is the depolarisation probability, swept over $p \in [0.00, 0.30]$ in steps of 0.01.

2. Mixed NISQ Noise Model

The primary validation experiment employs a composed noise channel:

$$\mathcal{E}_{\text{mix}} = \mathcal{E}_{\text{readout}} \circ \mathcal{E}_{\text{dephase}} \circ \mathcal{E}_{\text{amp}}, \quad (25)$$

where the components model amplitude damping ($T_1 \in [10, 100] \mu\text{s}$), dephasing ($T_2 \in [10, 100] \mu\text{s}$), and readout error ($p_{\text{readout}} \in [0.00, 0.05]$) respectively.

B. Noisy Swap Test and Comparison Reliability

For noisy states $\rho'_A = \mathcal{E}(\rho_A)$ and $\rho'_B = \mathcal{E}(\rho_B)$, we introduce a *comparison reliability metric*:

$$R_{\text{comp}} = \sqrt{\mathcal{F}_A \mathcal{F}_B} \cdot \sqrt{\gamma_{\perp,A} \gamma_{\perp,B}} \cdot \mathcal{S}. \quad (26)$$

Overlap scores may remain deceptively stable as both states drift toward similar mixed-state distributions under noise—the “thermal trap”—while R_{comp} deteriorates.

C. Variational Algorithms and Reliability-Aware Optimisation

1. Reliability-Adjusted Objective

The HDMF introduces a reliability-adjusted objective:

$$J_{\text{HDM}}(\theta) = J(\theta) \cdot R_{\text{HDM}}(\theta), \quad (27)$$

leading to the reliability-aware optimisation:

$$\theta^* = \arg \max_{\theta} J_{\text{HDM}}(\theta). \quad (28)$$

2. Reliability-Aware Stopping Criterion

A reliability-aware stopping criterion terminates optimisation when:

$$\frac{\Delta J}{\Delta t} > 0 \quad \text{but} \quad \frac{\Delta R_{\text{HDM}}}{\Delta t} < -\delta_R, \quad (29)$$

for some tolerance $\delta_R > 0$, preventing noise-induced over-optimisation into fragile regions of parameter space.

VII. RESULTS AND EMPIRICAL VALIDATION

A. Purpose and Design

Testing proceeded through five simulation stages, all evaluated against the Bell target state $|\Phi^+\rangle$, using both pure depolarising baseline models and mixed NISQ noise models. All simulations were performed using Qiskit [8] with full state tomography ($N_{\text{shots}} = 10,000$ per configuration).

B. Test 1: Divergence of Fidelity and Reliability

TABLE III. Fidelity and HDM reliability at increasing noise levels.

Noise parameter	Fidelity ($\mathcal{F}$)	HDM Reliability (R_{HDM})
0.10	0.72	0.32
0.20	0.54	0.13
0.30	0.42	0.07

R_{HDM} is strictly more conservative than $\mathcal{F}$ under mixed NISQ noise, penalising states that maintain geometric proximity to the target through favourable random noise rather than coherent quantum operations.

C. Test 2: Empirical Observation of the Hidden Reliability Region

Searching 10,000 noise configurations for states satisfying $\mathcal{F} \geq 0.90$ and $R_{\text{HDM}} < 0.60$, the simulation identified

190 distinct cases. A representative checkpoint yielded $\mathcal{F} \approx 0.95$ alongside $R_{\text{HDM}} \approx 0.50$.

D. Test 3: Predictive Utility for Downstream Computation

TABLE IV. Correlation of checkpoint metrics with downstream fidelity.

Metric	Pearson r
Checkpoint fidelity ($\mathcal{F}$)	0.02 ($p = 0.48$)
Checkpoint HDM reliability (R_{HDM})	0.9717 ($p < 10^{-6}$)

E. Test 4: Boundary Conditions

In environments bottlenecked exclusively by coherence loss, purity alone outperforms R_{HDM} . **Critical finding:** R_{HDM} is *not* universally dominant; users should not employ the HDMF when the noise landscape is known to be strictly one-dimensional.

F. Test 5: Multi-Factor Reliability Scaling

Under balanced multi-factor NISQ noise, R_{HDM} ranked first in predictive power, identifying **347** Hidden Reliability cases across 10,000 runs.

G. Summary

TABLE V. Summary of simulation runs and Hidden Reliability cases.

Test	Purpose	Runs	HRR Cases
1	Fidelity-reliability divergence	1,000	–
2	HRR observation	10,000	190
3	Downstream predictive utility	5,000	–
4	Boundary conditions	5,000	–
5	Multi-factor scaling	10,000	347
Total		31,000	537

VIII. LIMITATIONS AND CRITICAL EVALUATION

Correlated noise.: $\mathcal{S} = \prod_i (1 - p_i)$ assumes independent failures; correlated errors require an extended model.

One-dimensional dominance.: A single-mechanism noise environment is better served by the corresponding isolated metric.

Model dependence of $\gamma_{\perp}$.: Equation (16) is derived for the depolarising channel; recalculate $\hat{\gamma}(\mathcal{F})$ for other noise models.

Incomplete characterisation.: R_{HDM} inherits errors from an incomplete noise layer $\mathcal{N}$.

Simulation-only validation.: All results are from 2-qubit classical simulation; physical hardware validation (≥ 5 qubits) is required before deployment.

A. Recommendations

- Always inspect $\mathcal{F}$, $\gamma_{\perp}$, and $\mathcal{S}$ individually alongside R_{HDM} .
- Validate the independence assumption before deployment.
- Use the HDMF as a diagnostic *supplement*, not a replacement for established techniques.

IX. DISCUSSION

The HDMF shifts the central question from “What is the quantum state?” to “How much confidence should be placed in this computation?” The most striking empirical result is the near-perfect correlation ($r = 0.9717$, $p < 10^{-6}$) between R_{HDM} at a checkpoint and eventual downstream fidelity, versus $r \approx 0.02$ for checkpoint fidelity alone. The introduction of fidelity-residual purity $\gamma_{\perp}$ resolves the information-overlap problem without sacrificing the weakest-link interpretability of the multiplicative form. The reliability-aware stopping criterion (Eq. (29)) is the most pressing target for physical hardware testing.

X. FUTURE DIRECTIONS

Hardware validation.: Testing on physical NISQ devices (≥ 5 qubits), verifying HRR predictions.

Information-theoretic grounding.: Connecting R_{HDM} to quantum Fisher information or resource theories of coherence [7].

Correlated noise extension.: Extending $\mathcal{S}$ using crosstalk tensors or process matrix formalisms.

Multi-scale reliability.: Reliability at state, circuit, algorithm, and system levels.

Reliability-aware compilation.: Circuit routing and scheduling optimised for R_{HDM} rather than fidelity.

XI. CONCLUSION

We have introduced the Hybrid Density Matrix Framework, formalised as $\mathcal{H}_{\rho} = (\rho, \mathcal{N}, \mathcal{U}, \mathcal{C}, \mathcal{M})$ with explicit layer semantics and update rules, and realised through the decoupled reliability metric $R_{\text{HDM}} = \mathcal{F} \cdot \gamma_{\perp} \cdot \mathcal{S}$. The Hidden Reliability Region—where high fidelity masks low operational trustworthiness—is formally defined and empirically demonstrated across 537 of 31,000 simulated configurations. R_{HDM} predicts downstream performance with $r = 0.9717$ versus $r \approx 0$ for fidelity alone. The HDMF provides an integration architecture through which diverse reliability signals may be combined into a unified assessment of computational trustworthiness that fidelity alone cannot provide.

Appendix A: Simulation Parameters

TABLE VI. Noise model parameters.

Model	Parameters
Depolarising	$p \in [0.00, 0.30]$, step 0.01
Amplitude Damping	$T_1 \in [10, 100] \mu\text{s}$
Dephasing	$T_2 \in [10, 100] \mu\text{s}$
Readout Error	$p_{\text{readout}} \in [0.00, 0.05]$

TABLE VII. Circuit parameters.

Parameter	Value
Number of qubits	2
Circuit depth	1–10 layers
Number of shots	10,000
Tomography	Full state tomography

Appendix B: Code Availability

The simulation code is available at:

<https://github.com/BlakMajik/Hybrid-Density-Matrix-Framework>

Dependencies: Qiskit, NumPy, SciPy, Matplotlib.

ACKNOWLEDGMENTS

The author acknowledges the broader quantum computing community for developing the foundational techniques upon which this framework is built.

-
- [1] J. Preskill, “Quantum Computing in the NISQ era and beyond,” *Quantum* **2**, 79 (2018).
 - [2] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information* (Cambridge University Press, Cambridge, 2000).
 - [3] E. Knill *et al.*, “Randomized benchmarking of quantum gates,” *Phys. Rev. A* **77**, 012307 (2008).
 - [4] J. J. Wallman and J. Emerson, “Noise tailoring for scalable quantum computation via randomized compiling,” *Phys. Rev. A* **94**, 052325 (2016).
 - [5] S. Endo *et al.*, “Quantum error mitigation,” *J. Phys. Soc. Jpn.* **90**, 032001 (2021).
 - [6] R. Blume-Kohout, “Optimal, reliable estimation of quantum states,” *New J. Phys.* **12**, 043034 (2010).
 - [7] A. Winter and D. Yang, “Operational resource theory of coherence,” *Phys. Rev. Lett.* **116**, 120404 (2016).
 - [8] Qiskit contributors, “Qiskit: An open-source framework for quantum computing,” (2023), <https://doi.org/10.5281/zenodo.2573505>.